

ENVS 193 Midterm

Kate Bierwagen

2026-05-17

```
# loading in packages
library(tidyverse)
library(janitor)
library(rstatix)
library(effectsize)
library(gt)
```

```
# reading in data
halibut <- read_csv("halibut.csv")
bee_pathogens <- read_csv("bee-pathogens.csv")
```

Problem 1. Understanding, critiquing, and adding to written communication

a.

- This test could either be a two-sample t test or a Mann-Whitney U test because both of those tests are comparing two independent groups, commercial halibut landings between Southern California and Northern California.
- Both these tests compare two independent groups, but differ with what it compares.
- The two-sample t test compares means and is used when data is normally distributed. The Mann-Whitney U test compares medians or rank distributions and is a non-parametric test.

b. Writing new statements

- two-sample t-test: No significant difference between commercial halibut landings (lbs) between Northern California (mean = $\bar{x}$, $n = n$) and Southern California (mean = $\bar{x}$, n

= n) across years (1969-2019) (two-sample t-test: $t(\text{degrees of freedom}) = t\text{-statistic}$, $p = 0.3$, $\alpha = 0.05$).

- Mann-Whitney U test:

No significant difference between commercial halibut landings (lbs) between Northern California (median = $\tilde{x}$, $n = n$) and Southern California (median = $\tilde{x}$, $n = n$) across years (1969-2019) (Mann-Whitney, $W = u\text{-statistic}$, $p = 0.3$, $\alpha = 0.05$).

c.New figures

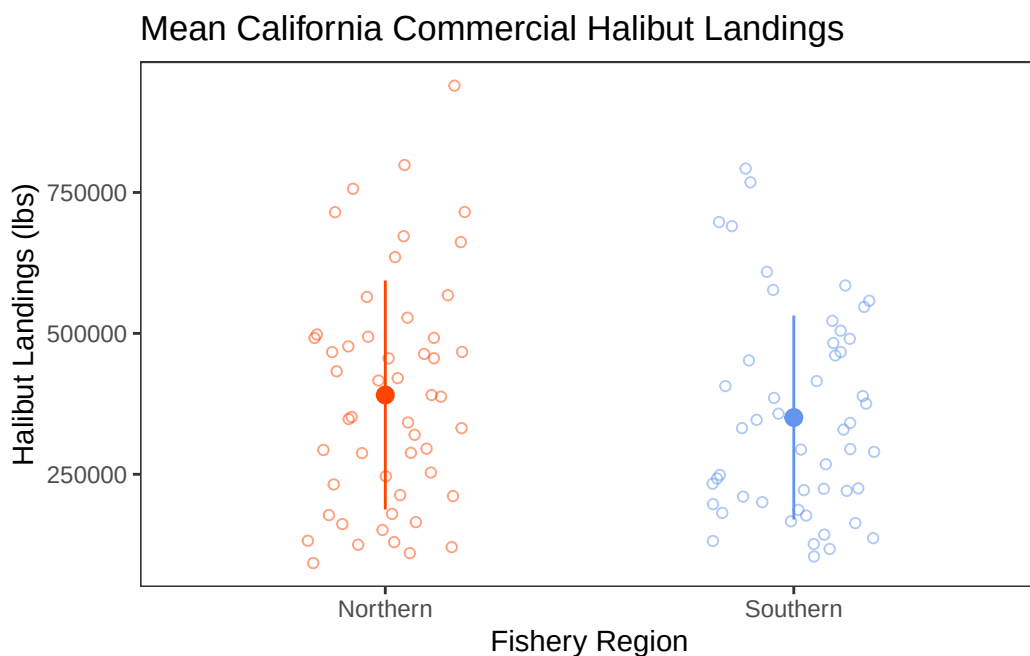
```
# calculating central tendencies
halibut_summary <- halibut|>
# grouping by regional landing
group_by(ca_region) |>
# summarizing by landing weight
summarize(
  mean = mean(landings_lb), # mean, central tendency
  n = length(landings_lb), # number of observations
  sd = sd(landings_lb), # standard deviation
  median = median(landings_lb), # median, central tendency
  df = n - 1) |> # degrees of freedom
# ungrouping data frame
ungroup()

# base layer: ggplot, using halibut data frame
ggplot(data = halibut,
  mapping = aes(x = ca_region, # x axis: fishery region
    y = landings_lb, # y axis: halibut landings
    color = ca_region)) + # coloring by region
# first layer: showing underlying data
geom_jitter(height = 0, # no jitter in vertical direction
  alpha = 0.5, # making the points transparent
  shape = 1, # making points to open circles
  width = 0.2) + # narrowing jitter
# second layer: displaying mean and standard deviation
geom_pointrange(data = halibut_summary,
  mapping = aes(x = ca_region, # x- axis
    y = mean, # y- axis
    ymin = mean - sd, # upper bound sd
    ymax = mean + sd)) + # lower bound sd
# changing site colors from the default
```

```

scale_colour_manual(values = c("Northern" = "orangered",
                                "Southern" = "cornflowerblue")) +
# relabeling x- and y- axes and adding title
labs(title = "Mean California Commercial Halibut Landings",
      x = "Fishery Region",
      y = "Halibut Landings (lbs)") +
# changing theme from default
theme_bw() +
# getting rid of legend and grid lines
theme(legend.position = "none",
      panel.grid = element_blank())

```



```

# base layer: ggplot using halibut data frame
ggplot(data = halibut,
       mapping = aes(x = ca_region, # x axis: fishery region
                     y = landings_lb, # y axis: halibut landings
                     color = ca_region)) + # coloring by fishery region
# first layer: boxplot
geom_boxplot() +
# second layer: showing underlying data
geom_jitter(height = 0, # making jitter vertically
            width = 0.2, # smaller jitter in horizontal direction)

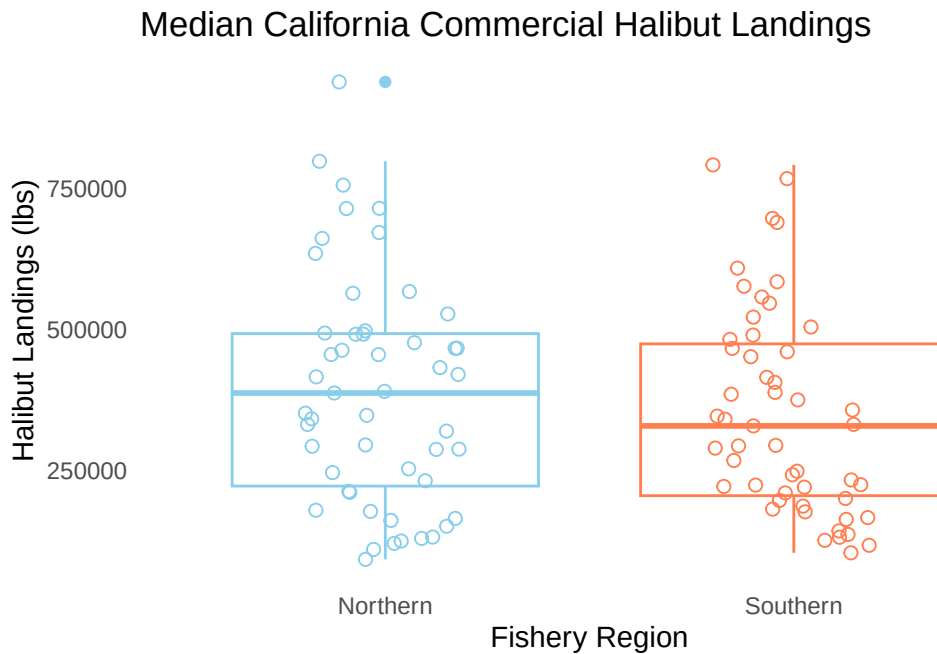
```

```

    shape = 1, # making points to open circles
    size = 2) + # changing shape to closed circles
# adding title and relabeling x- and y- axes
labs(title = "Median California Commercial Halibut Landings",
     x = "Fishery Region",
     y = "Halibut Landings (lbs)") +
# changing colors of locations from default
scale_color_manual(values = c("Northern" = "skyblue",
                              "Southern" = "coral")) +

# changing theme from default
theme_minimal() +
# removing legend and grid lines
theme(legend.position = "none",
      panel.grid = element_blank())

```



d. Caption Writing

- Mean figure: Figure 1. Mean commercial halibut landings (lbs) by fishery region in California. Point and error bars represent mean and standard deviation of halibut landings for Northern and Southern California fishery regions. Open circles represent individual commercial halibut landings (lbs) to show the underlying data. Data from California

Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

- Median Figure: Figure 2. Median commercial halibut landings (lbs) by fishery region in California. Boxplots represent the median and interquartile range of halibut landings (lbs) for Northern and Southern California fishery regions. Open circles represent individual halibut landings (lbs) to show underlying distribution. Data from California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 2. Data visualization

a. Plan your figure

- x-axis (name the specific column): **year**
- y-axis (name the specific column): **landings_lb**
- geometry function (hint: there is only one): **geom_line()**
- colors and linetypes (name the specific column): **ca_region**
- relabelling the x- and y-axes (name the specific function): **labs()**
- using custom colors (name the specific function): **scale_color_manual()**
- theme arguments and the `element_` for:
 - positioning the x-axis text at a 45° angle: **theme(axis.title.x = element_text(angle = 45))**
 - putting the legend in the top right of the plot: **theme(legend.position = c(0.85, 0.85))**
 - removing the panel background: **theme(panel.background = element_blank())**
 - removing the panel grid: **theme(panel.grid = element_blank())**
 - removing the axis ticks: **theme(axis.ticks = element_blank())**
 - removing the legend title: **theme(legend.title = "none")**

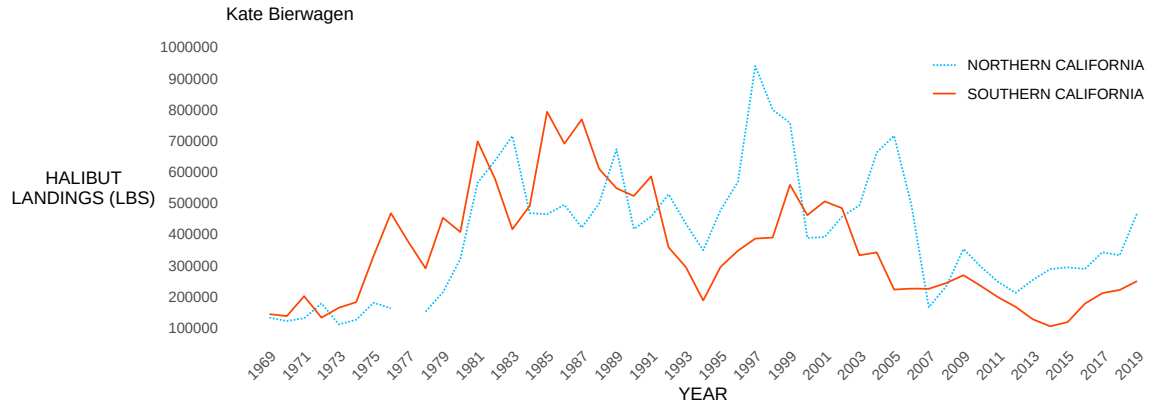
b. Code your figure

```
# base layer: ggplot using halibut data frame
ggplot(data = halibut,
  # x-axis: year (time)
  # y-axis: metric tons in millions
  # color by production type
  # make lines by production type
  mapping = aes(x = year,
```

```

        y = landings_lb,
        color = ca_region,
        linetype = ca_region)) +
# second layer: lines
geom_line() +
# custom colors, labeling the lines
scale_color_manual(
  values = c("Northern" = "deepskyblue",
             "Southern" = "orangered"),
  labels = c("NORTHERN CALIFORNIA",
             "SOUTHERN CALIFORNIA"))+
# making northern region dotted line and southern region solid line
scale_linetype_manual(values = c("11", "solid"),
                      labels = c("NORTHERN CALIFORNIA",
                                "SOUTHERN CALIFORNIA")) +
# making years show every other year until 2019
scale_x_continuous(breaks = seq(1969, 2019, by = 2)) +
# making y axis range 10000-100000 and values increase by 100000
scale_y_continuous(limits = c(100000, 1000000),
                   breaks = seq(100000, 1000000, by = 100000),
                   labels = scales::label_number(big.mark = "")) +
# relabelling x- and y- axes
labs(x = "YEAR",
     y = "HALIBUT\nLANDINGS (LBS)", # splitting y axis label into two lines
     color = "Production type", # coloring lines
     linetype = "Production type", # labeling lines
     subtitle = "Kate Bierwagen") + # putting name as subtitle
# custom theme
theme_minimal() +
# changing y axis title to be horizontal and x- axis to be at 45°
theme(axis.text.x = element_text(angle = 45, hjust = 1), # right aligned
      axis.title.y = element_text(angle = 0, vjust = 0.5), # centered
      legend.position = c(0.85, 0.85), # making legend top right
      panel.background = element_blank(), # removing panel background
      panel.grid = element_blank(), # removing panel grid
      axis.ticks = element_blank(), # removing axis ticks
      legend.title = element_blank()) # removing legend title

```



c. Write a caption

Figure 1. Halibut landings (lbs) over time between Northern and Southern California fishery regions. Lines represent yearly halibut landings from 1969-2019 by color (red line: Southern California, blue line: Northern California). Data from California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 3. Reproducing an analysis

a. Variables

- The response variable is Black queen cell virus loads in bees, which is a continuous quantitative variable.
- The predictor variable is habitat type in Pennsylvania: valley, edge, and forest.
- The researchers measured Black queen cell virus loads by performing a quantitative PCR (qPCR) test on the pooled abdomens of 3 *B. impatiens* workers. This is described in Paragraphs 3 and 4 under Pathogen Quantification in the methods section

b. Test Type

The researchers performed a Kruskal-Wallis test, which was followed by a Dunn post-hoc test among habitat types with a Bonferroni correction. This is found in paragraph 11 of the methods section under statistical analysis.

c. Hypotheses

H_0 : Median BQCV loads between valley, edge, and forest habitat types are all equal to each other.

H_A : At least two habitat types (valley, edge, and forest) have different median BQCV loads from each other.

d. Cleaning and organizing data

```
# creating new object to clean and wrangle data
bee_pathogens_clean <- bee_pathogens |>
  # cleaning column names
  clean_names() |>
  # selecting only columns representing BQCV and habitat/site
  select(site_type_x,
         bqcv) |>
  # renaming site column to habitat
  rename(habitat = site_type_x) |>
  # dropping values in BQCV column that are NA
  drop_na(bqcv)

# using slice sample to display 1- rows of the bee_pathogens_clean data frame
bee_pathogens_clean |> slice_sample(n = 10)
```

```
# A tibble: 10 x 2
  habitat    bqcv
  <chr>    <dbl>
1 edge      1.60
2 edge      9.42
3 edge      4.68
4 edge      8.41
5 valley    8.42
6 valley    9.70
7 edge      9.64
8 edge      8.92
9 valley    7.56
10 valley    5.22
```

e. Summarizing and table displays

```
# creating object from bee_pathogens_clean to summarize data
bee_pathogens_summary <- bee_pathogens_clean |>
  # grouping by habitat type
  group_by(habitat) |>
  # calculating summary statistics
  summarize(
    median = round(median(bqcv), 1), # median, rounding to 1 decimal place
    n = length(bqcv)) |> # calculating sample sizes
  # ungrouping by habitat
  ungroup()
```

```
bee_pathogens_summary |>
  # making table using gt
  gt() |>
  # changing column names
  cols_label(
    habitat = "Habitat Type",
    n = "Sample Size (n)",
    median = "Median BQCV Load") |>
  tab_options(latex.tbl.pos = "h", # h for render the table "here"
    latex.use_longtable = TRUE, # setting PDF rendering option
    table.width = px(500)) # making the table wider)
```

Habitat Type	Median BQCV Load	Sample Size (n)
edge	6.3	27
forest	5.0	24
valley	7.6	29

f. Do a test

```
# performing a kruskal- wallis test
kruskal.test(
  # formula: bqcv load as a function of habitat type
  bqcv ~ habitat,
  # data frame: clean bee pathogens data frame
  data = bee_pathogens_clean)
```

Kruskal-Wallis rank sum test

data: bqcv by habitat

Kruskal-Wallis chi-squared = 9.8219, df = 2, p-value = 0.007365

g. Do a post-hoc test.

```
# performing a post-hoc test
dunn_test(
  # formula: bqcv load as a function of habitat type
  bqcv ~ habitat,
  # data frame: clean names bee pathogens data frame
  data = bee_pathogens_clean,

  p.adjust.method = "bonferroni")
```

```
# A tibble: 3 x 9
  .y. group1 group2  n1  n2 statistic      p  p.adj p.adj.signif
* <chr> <chr>  <chr> <int> <int>    <dbl>  <dbl>  <dbl> <chr>
1 bqcv  edge    forest   27   24    -1.83 0.0679  0.204  ns
2 bqcv  edge    valley   27   29     1.31 0.189   0.567  ns
3 bqcv  forest  valley   24   29     3.13 0.00175 0.00525 **
```

h. Calculate an effect size.

```
# calculating effect size
kruskal_effsize(
  # formula: bqcv load as a function of habitat type
  bqcv ~ habitat,
  # data frame: clean names bee pathogens data frame
  data = bee_pathogens_clean)
```

```
# A tibble: 1 x 5
  .y.      n effsize method magnitude
* <chr> <int>  <dbl> <chr>  <ord>
1 bqcv   80   0.102 eta2[H] moderate
```

i. Written communication

We found a moderate ($\eta^2 = 0.102$) difference in BQCV loads between forest, edge, and valley habitats types (Kruskal-Wallis test; $\chi^2(2) = 9.82$, $p = 0.007$, $\alpha = 0.05$), with significant differences between valley (median: 7.6, $n = 29$) and forest (median = 5.0, $n = 24$, $z = -3.13$, Dunn's post-hoc test $p\text{-adj} = 0.005$), but edge (median = 6.3, $n = 27$) was not different from either habitat type, valley (Dunn's post-hoc test, $p\text{-adj} = 0.189$) and forest (Dunn's post-hoc test, $p\text{-adj} = 0.204$).